EVENT OVERVIEW

Select one earthquake. Counts use the current building stock and 2023 population (as-if-today).

Event

Kaikōura, 2016

SOURCE AND BASIS

Event	Shaking Source	Exposure Basis
Kaikōura, 2016	official	As-if-today

MMI Threshold

6

MAGNITUDE

7.8

MAXIMUM MMI

8.4

EXPOSED BUILDINGS

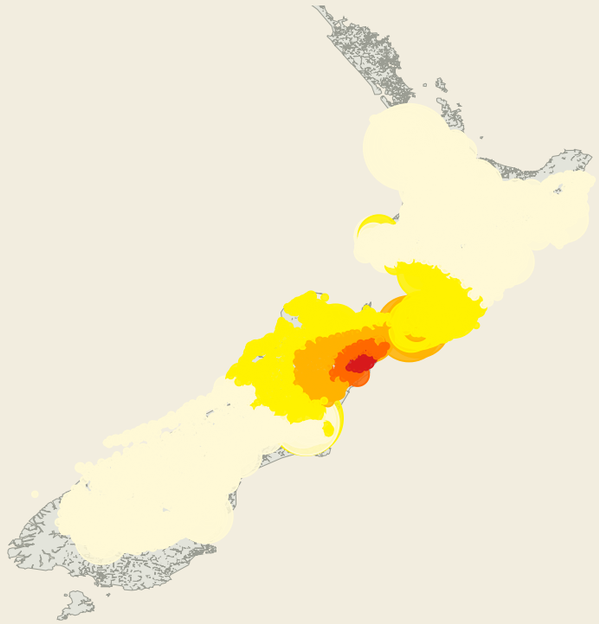
78,564

EXPOSED PEOPLE

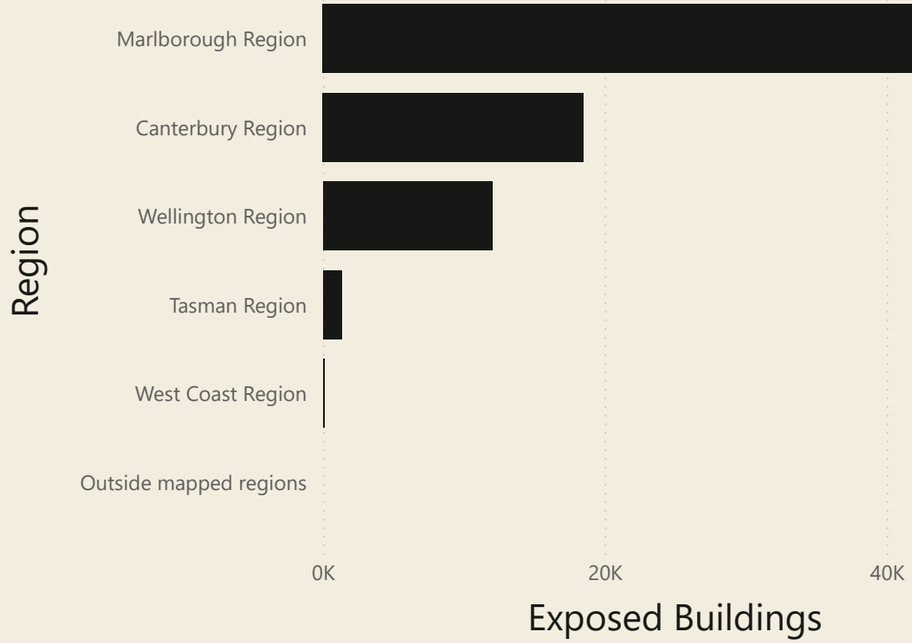
83,276

H3 EXPOSURE POINTS / MMI INTENSITY

MMI Band MMI 4 MMI 5 MMI 6 MMI 7 MMI 8+



REGIONS BY EXPOSED BUILDINGS



NZ HAZARD EXPOSURE MONITOR / AS-IF-TODAY REPLAY / 2023 POPULATION

EVENT COMPARISON

Ten fixed earthquakes replayed against today's buildings and the 2023 population.

MMI Threshold

6

HISTORICAL EVENTS

10

LARGEST EVENT EXPOSURE

337,971

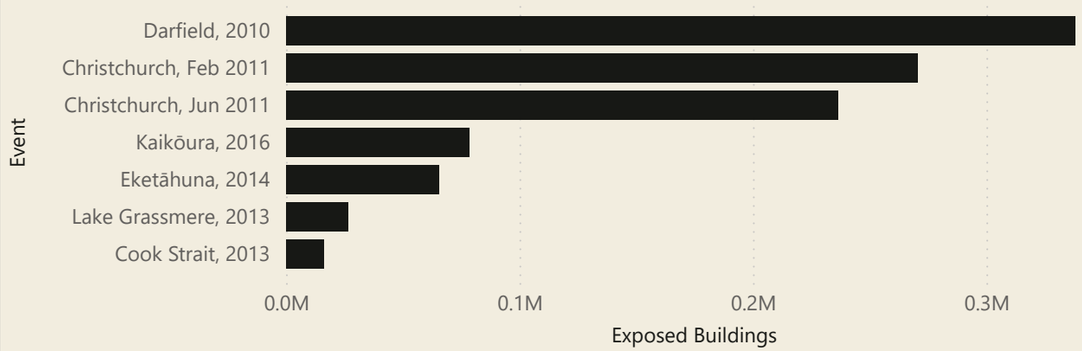
EVENT WITH LARGEST EXPOSURE

Darfield, 2010

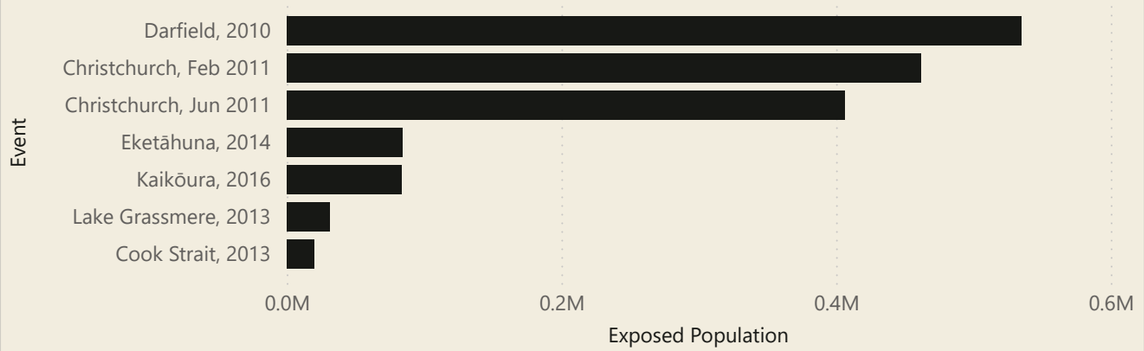
EVENTS WITH DAMAGING SHAKING

7

BUILDINGS / EVENT



PEOPLE / EVENT



EVENT COMPARISON / SELECT ANY ROW TO CROSS-FILTER

Event	Magnitude	Max MMI	Exposed Buildings	Exposed Population	
Darfield, 2010	7.2	8.4	337,971	534,594	
Christchurch, Feb 2011	6.2	9.4	270,518	461,765	
Christchurch, Jun 2011	6.0	8.8	236,306	406,499	
Kaikōura, 2016	7.8	8.4	78,564	83,276	
Eketāhuna, 2014	6.2	6.8	65,461	84,416	
Lake Grassmere, 2013	6.5	8.0	26,492	31,464	
Cook Strait, 2013	6.5	7.6	16,059	19,950	
East Cape, 2016	7.1	5.2			
Kermadec Islands, 2021	7.9	2.2			
Taumarunui, Oct 2026	5.7	2.8			

NZ HAZARD EXPOSURE MONITOR / AS-IF-TODAY REPLAY / 2023 POPULATION

REGIONAL EXPOSURE

Cell-to-region attribution follows the majority building location; boundaries are approximate.

Event

Kaikōura, 2016

MMI Threshold

6

EXPOSED BUILDINGS

78,564

Share of national stock

2.4%

REPEAT EXPOSURE ACROSS 10 REPLAYS

1,031,371

BUILDINGS IN MMI 6+ CELLS

491,689

REPEAT EXPOSURE ACROSS 10 REPLAYS / CENTROIDS



REGION / DISTRICT RANKING AND REPEAT EXPOSURE

Region	Exposed Buildings	% Region Buildings Exposed	Accumulated Building Exposures
Marlborough Region	46,691	87.4%	79,965
Canterbury Region	18,496	3.9%	863,320
Wellington Region	12,031	4.3%	23,783
Tasman Region	1,344	2.6%	1,344
West Coast Region	2	0.0%	2
Outside mapped regions	0		0
Auckland Region			0
Bay of Plenty Region			0
Gisborne Region			0
Hawke's Bay Region			0
Manawatū-Whanganui Region			62,957
Nelson Region			0
Northland Region			0
Total	**78,564**	**2.4%**	**1,031,371**

NZ HAZARD EXPOSURE MONITOR / AS-IF-TODAY REPLAY / 2023 POPULATION

EVENT HISTORY

Event time along the horizontal axis, magnitude vertically, bubble size by thresholded building exposure.

Region

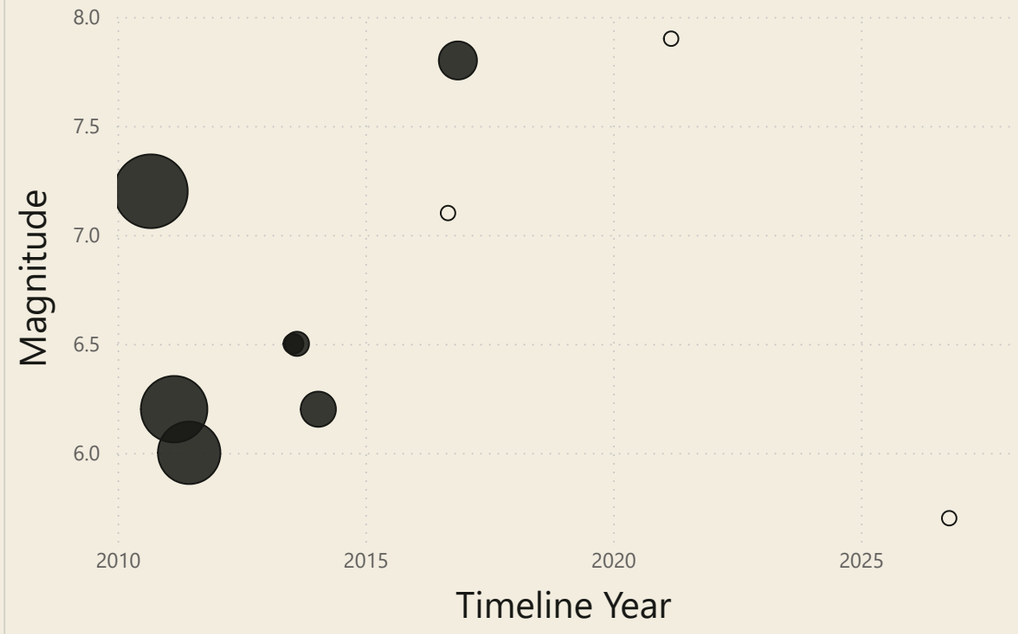
- [ ] Auckland Region
- [ ] Bay of Plenty Region

MMI Threshold

6

EVENTS	EVENTS WITH DAMAGING SHAKING	LARGEST EXPOSURE EVENT	LARGEST EVENT EXPOSURE
10	7	Darfield, 2010	337,971

EVENT TIME × MAGNITUDE / BUBBLE AREA = EXPOSED BUILDINGS



EVENT CHRONOLOGY

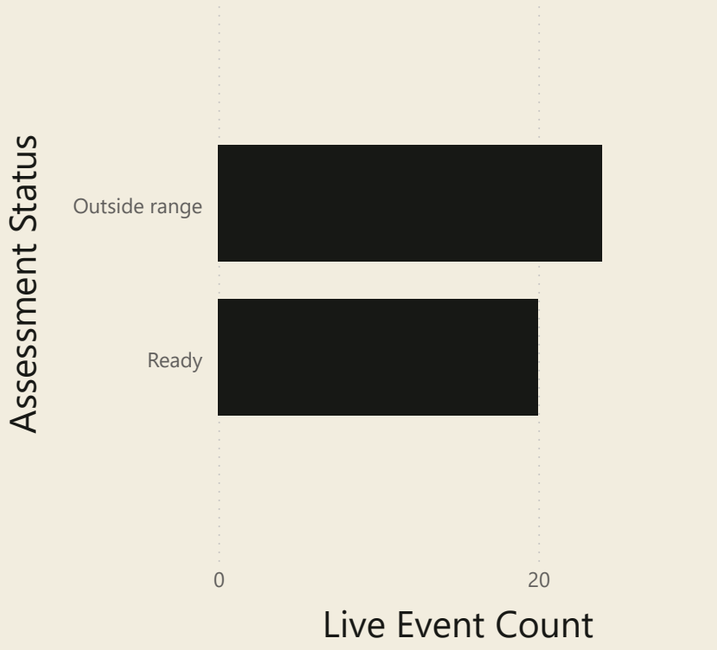
Event	Origin Time NZ	Magnitude	Exposed Buildings
Darfield, 2010	2010-09-04 04:35 NZST	7.2	337,971
Christchurch, Feb 2011	2011-02-22 12:51 NZDT	6.2	270,518
Christchurch, Jun 2011	2011-06-13 14:20 NZST	6.0	236,306
Cook Strait, 2013	2013-07-21 17:09 NZST	6.5	16,059
Lake Grassmere, 2013	2013-08-16 14:31 NZST	6.5	26,492
Eketāhuna, 2014	2014-01-20 15:52 NZDT	6.2	65,461
East Cape, 2016	2016-09-02 04:37 NZST	7.1	
Kaikōura, 2016	2016-11-14 00:02 NZDT	7.8	78,564
Kermadec Islands, 2021	2021-03-05 08:28 NZDT	7.9	
Taumarunui, Oct 2026	2026-10-15 18:25 NZDT	5.7	

DATA HEALTH

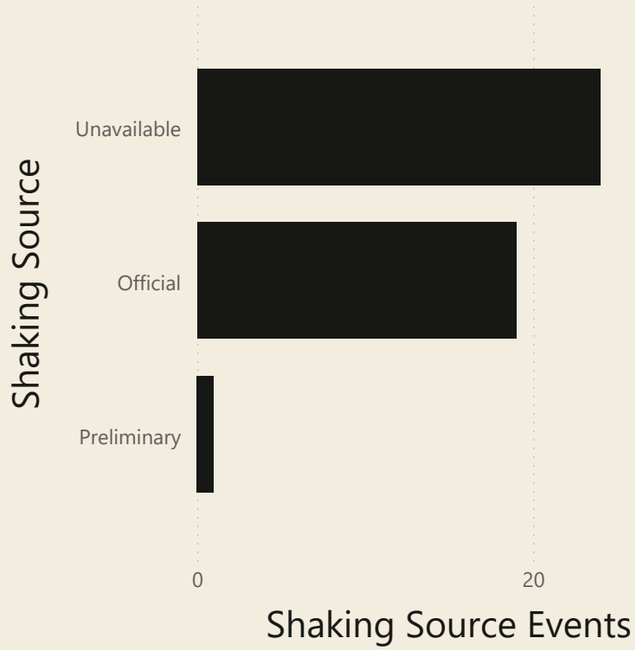
Azure gold and runtime, read-only at refresh time; not a live dashboard.

LAST SUCCESSFUL RUN / UTC	HOURS SINCE LAST RUN	FAILED RUNS	POISON MESSAGES	STATION-WEIGHTED MAE
5 Nov 2026 00:00	11	0	27	1.049

RECENT EVENT STATUS



RECENT SHAKING SOURCE



MODEL VALIDATION BY EVENT

Event	Validation Stations	Estimated MMI MAE
Kermadec Islands, 2021	1	2.821
Taumarunui, Oct 2026	181	2.309
Kaikōura, 2016	323	1.373
Darfield, 2010	425	1.160
East Cape, 2016	151	1.136
Christchurch, Feb 2011	201	0.861
Cook Strait, 2013	360	0.852
Christchurch, Jun 2011	193	0.849
Lake Grassmere, 2013	289	0.787
Eketāhuna, 2014	423	0.641

READ-ONLY AZURE SNAPSHOT. Housekeeping failures and poison-queue depth are captured at refresh time. Recent event status and shaking source come from the published index. The station-weighted MAE is indicative, not an actuarial loss estimate. Historical replays use current building outlines and 2023 population. Water cells and the Chatham Islands are outside the mapped exposure model.